

Student Performance Dataset

Pandas + Matplotlib + Seaborn Operations with Code

Use the file **student_performance.csv**. Run the following operations in Jupyter Notebook, Google Colab, or VS Code.

1. Import Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

2. Read the CSV File

```
df = pd.read_csv("student_performance.csv")
print(df)
```

3. Display First 5 Rows

```
df.head()
```

4. Display Last 5 Rows

```
df.tail()
```

5. Check Rows and Columns

```
df.shape
```

6. Display Column Names

```
df.columns
```

7. Check Data Types

```
df.dtypes
```

8. Check Missing Values

```
df.isnull().sum()
```

9. Statistical Summary

```
df.describe()
```

10. Find Average Final Score

```
df["Final_Score"].mean()
```

11. Find Highest Final Score

```
df["Final_Score"].max()
```

12. Find Lowest Final Score

```
df["Final_Score"].min()
```

13. Find Top-Scoring Student

```
df.loc[df["Final_Score"].idxmax()]
```

14. Students with Final Score > 80

```
df[df["Final_Score"] > 80]
```

15. Students with Study Hours > 6

```
df[df["Study_Hours"] > 6]
```

16. Students with Attendance > 90

```
df[df["Attendance"] > 90]
```

17. Sort by Final Score

```
df.sort_values("Final_Score", ascending=False)
```

18. Average Score by Gender

```
df.groupby("Gender")["Final_Score"].mean()
```

19. Average Marks in Each Subject

```
df[["Maths", "Science", "English"]].mean()
```

20. Students Scoring > 80 in All Subjects

```
df[(df["Maths"] > 80) &  
    (df["Science"] > 80) &  
    (df["English"] > 80)]
```

21. Matplotlib – Bar Chart

```
plt.bar(df["Name"], df["Final_Score"])  
plt.xlabel("Student")  
plt.ylabel("Final Score")  
plt.title("Student Final Scores")  
plt.xticks(rotation=90)  
plt.show()
```

22. Matplotlib – Line Plot

```
plt.plot(df["Study_Hours"], df["Final_Score"], marker="o")  
plt.xlabel("Study Hours")  
plt.ylabel("Final Score")  
plt.title("Study Hours vs Final Score")  
plt.grid()  
plt.show()
```

23. Matplotlib – Scatter Plot

```
plt.scatter(df["Study_Hours"], df["Final_Score"])  
plt.xlabel("Study Hours")  
plt.ylabel("Final Score")  
plt.title("Study Hours vs Final Score")  
plt.show()
```

24. Matplotlib – Histogram

```
plt.hist(df["Final_Score"], bins=5)  
plt.xlabel("Final Score")  
plt.ylabel("Number of Students")
```

```
plt.title("Distribution of Final Scores")
plt.show()
```

25. Matplotlib – Pie Chart

```
gender_count = df["Gender"].value_counts()

plt.pie(gender_count, labels=gender_count.index, autopct="%1.1f%%")
plt.title("Gender Distribution")
plt.show()
```

26. Seaborn – Count Plot

```
sns.countplot(data=df, x="Gender")
plt.title("Number of Male and Female Students")
plt.show()
```

27. Seaborn – Bar Plot

```
sns.barplot(data=df, x="Gender", y="Final_Score")
plt.title("Average Final Score by Gender")
plt.show()
```

28. Seaborn – Scatter Plot with Hue

```
sns.scatterplot(
    data=df,
    x="Study_Hours",
    y="Final_Score",
    hue="Gender"
)
plt.title("Study Hours vs Final Score")
plt.show()
```

29. Seaborn – Box Plot

```
sns.boxplot(data=df[["Maths", "Science", "English"]])
plt.title("Subject Marks Distribution")
plt.show()
```

30. Seaborn – Heatmap

```
correlation = df.select_dtypes("number").corr()

sns.heatmap(correlation, annot=True)
plt.title("Correlation Heatmap")
plt.show()
```

31. Seaborn – Pair Plot

```
sns.pairplot(
    df[["Study_Hours", "Attendance", "Final_Score"]]
)
plt.show()
```

32. Save a Matplotlib Chart

```
plt.figure(figsize=(10, 5))
plt.bar(df["Name"], df["Final_Score"])
plt.xticks(rotation=90)
plt.title("Student Final Scores")
plt.savefig("student_scores.png")
plt.show()
```

33. Final Mini Practice

```
# Find the student with the highest Maths score
df.loc[df["Maths"].idxmax()]

# Find the student with the highest attendance
df.loc[df["Attendance"].idxmax()]

# Find average Final Score
df["Final_Score"].mean()

# Find correlation between Study Hours and Final Score
df["Study_Hours"].corr(df["Final_Score"])
```

Practice Questions

1. Does study time affect final score?
2. Does attendance affect final score?
3. Which subject has the highest average marks?
4. Which gender has the highest average final score?
5. Who is the top-performing student?
6. Which students scored above 80 in all three subjects?
7. What does the correlation heatmap tell you?